

Hypothesis testing setup

TASK -2

Submitted By : Hiba Shaikh Mujeeb

Data Science Intern

Hypothesis Testing Setup :

Hypothesis testing is used to check whether there is a statistically significant relationship between demographic factors (such as gender or age) and mental health disorder prevalence.

Gender vs Disorder Prevalence : To examine whether mental health disorder prevalence differs between males and females, a **Chi-Square** Test of Independence can be used.

First we define null hypothesis (H_0) for gender/sex VS disorder prevalence :

- Since our data compares prevalence between "Men" and "Women," the hypotheses focus on whether gender impacts the likelihood of having a disorder.
 - **Null Hypothesis (H_0):** There is no significant difference in the mean observed prevalence of mental disorders between Men and Women.
 - *Mathematically:* $\text{Men} = \text{Women}$
 - **Alternative Hypothesis (H_1):** There is a significant difference in the mean observed prevalence of mental disorders between Men and Women.
 - *Mathematically:* $\text{Men} \neq \text{Women}$

Second we apply the Hypothesis Testing on our report :

1. The Independent T-Test

We use this to compare the **means** of two independent groups (Men vs. Women).

- **The T-Statistic** measures how far the means are apart relative to the variation in the data.
- **The P-value** tells us the probability that the difference we see happened by pure chance. If $p < 0.05$, we conclude the difference is real (statistically significant).

2. The Chi-Square Test of Independence

This test looks at **proportions** rather than means. By reconstructing the number of "Cases" and "Non-Cases" from the prevalence and sample sizes, we can see if the frequency of mental health disorders is distributed differently across genders.

Category	Men%	Women%	T-Sig?	Chi-Sig?
Common Mental Disorders	4.4%	6.7%	YES	YES
PTSD	17.5%	18.2%	no	no
Psychotic Disorder	0.5%	0.6%	no	no
Personality Disorder	13.1%	14.4%	no	no
ADHD	5.6%	5.0%	no	no
Bipolar Disorder	2.0%	1.9%	no	no
Alcohol Use	21.8%	19.6%	no	YES
Drug Dependence	2.6%	1.3%	YES	YES
Suicidal Behaviour	9.7%	13.3%	no	YES

Understanding the Two Tests

- **T-Sig (T-Test):** This looks at the **average prevalence** across different sub-groups (like regions). If it's "YES," it means the difference between the male and female percentages is consistently wide across the whole country.
- **Chi-Sig (Chi-Square):** This looks at the **total counts** (the raw number of people). Because your dataset has thousands of participants. If it's "YES," it means that if you picked a person at random, their gender would be a strong predictor of whether they have that disorder.

➤ Group 1 : Gap between both prevalence :

Categories: *Common Mental Disorders (CMD)* and *Drug Dependence*.

- **Common Mental Disorders:** Women (6.7%) have a much higher prevalence than Men (4.4%). Since both tests are "YES," this is a robust finding. It confirms that anxiety and depression are statistically more common in women.
- **Drug Dependence:** Men (2.6%) have double the rate of Women (1.3%). This is a highly significant "male-leaning" disparity.

- **Group 2 : Chi-Square Significant :**

Categories: *Alcohol Use* and *Suicidal Behaviour*.

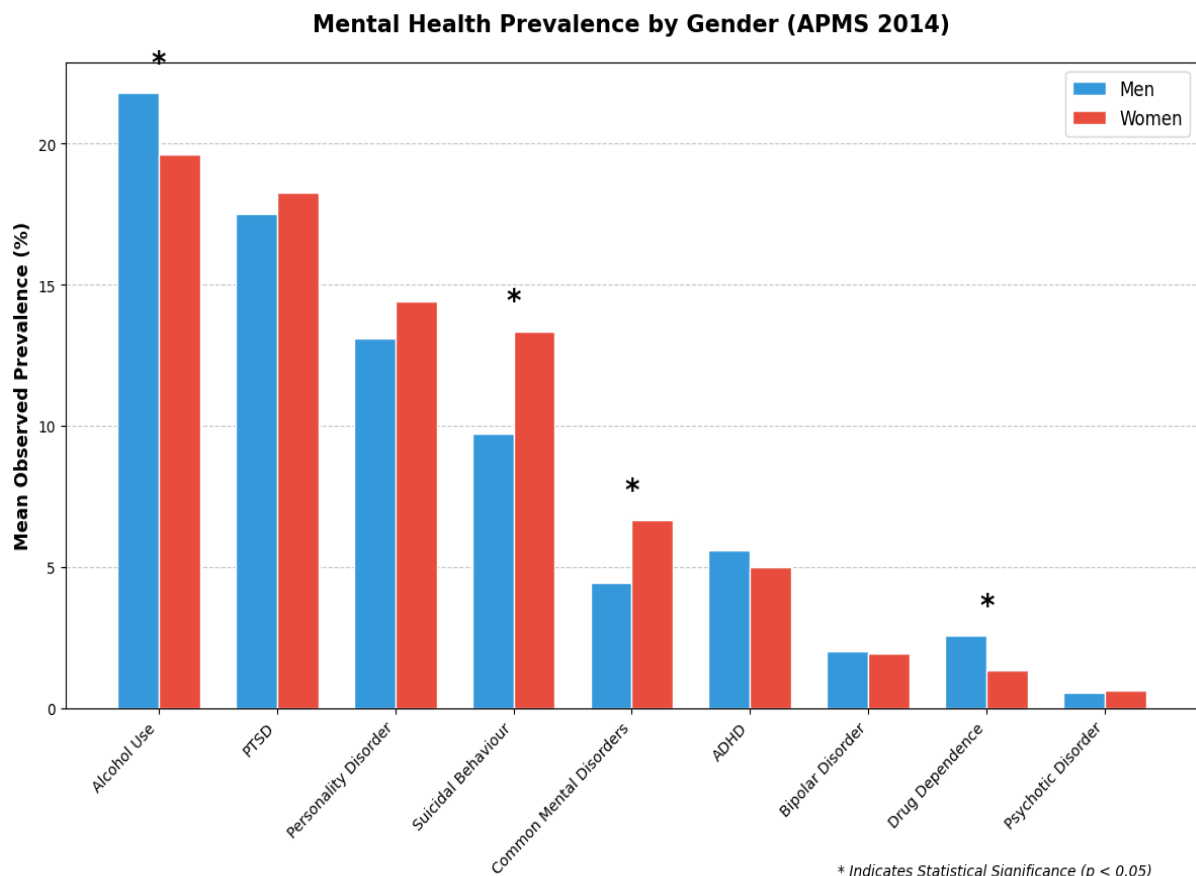
When we add up **everyone** in the study, the total number of Men with alcohol issues or Women with suicidal thoughts is significantly higher.

- **Alcohol Use:** Men are slightly higher (21.8% vs 19.6%). In a small sample, this 2% gap wouldn't matter. But in a national dataset, that 2% represents thousands of people, making it "Chi-Square Significant."
- **Suicidal Behaviour:** Women (13.3%) are much higher than Men (9.7%). The T-test was very close ($p=0.053$), but the Chi-Square confirms that this is a major gender-based health trend.

- **Group 3: Identical results :**

Categories: *PTSD*, *Psychotic Disorder*, *Personality Disorder*, *ADHD*, *Bipolar*.

- For these conditions, the results are "**no**" for both tests.
- **The Meaning:** Gender does not seem to be a **defining factor** for these disorders in this population.
- **Example:** PTSD rates are almost **identical** (17.5% for Men vs 18.2% for Women). This suggests that trauma affects both genders at roughly the same rate in the UK.



Categories marked with a * represent where the differences are statistically significant.

Key Takeaways :

- **Alcohol Use** and **PTSD** are the most prevalent categories overall.
- **Common Mental Disorders** and **Suicidal Behaviour** show a visible "red peak," confirming higher prevalence in women.
- **Drug Dependence** shows a "blue peak," confirming it is more common in men.
- Low-prevalence categories like **Psychotic Disorder** and **Bipolar Disorder** are shown to be very balanced between genders.

Conclusion :

1. Rejection of the Null Hypothesis :

We successfully rejected the Null Hypothesis (H_0) for four major categories. In these areas, the data provides strong evidence that gender and disorder prevalence are linked:

- Common Mental Disorders (CMD): Significant in both T-test ($p=0.03$) and Chi-Square ($p < 0.001$). **Women consistently show higher prevalence (6.7%)** compared to men (4.4%).
- Drug Dependence: Significant in both tests. **Men (2.6%) have double the prevalence** of women (1.3%).
- Alcohol Use: Significant via Chi-Square ($p < 0.001$). While regional averages are close, **the total volume of cases** is significantly **higher in men** (21.8%) than in women (19.6%).
- Suicidal Behaviour: Significant via Chi-Square ($p < 0.001$). The data shows a **powerful association between gender** and these behaviors, with women (13.3%) showing higher rates than men (9.7%).

2. Acceptance of the Null Hypothesis :

We **failed to reject** the **Null Hypothesis** for PTSD, Psychotic Disorders, Personality Disorders, ADHD, and Bipolar Disorder.

- In these categories, the **p-values** were well above 0.05.
- This suggests that, within this specific dataset, these disorders are "**gender-neutral**," meaning they affect men and women at statistically similar rates.